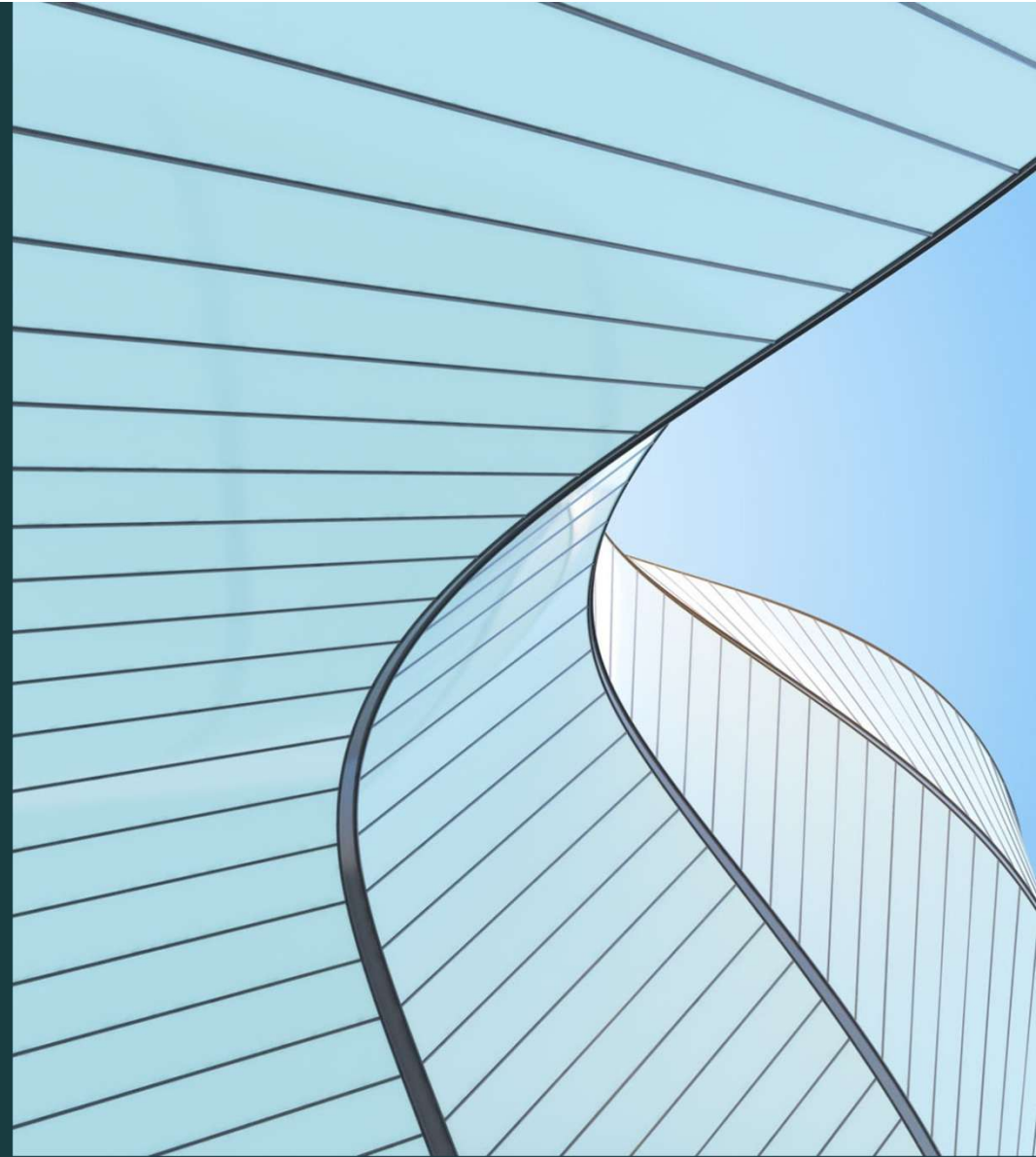


Team Caterpillar
10/15

ARINC 429 Midterm Presentation

This presentation outlines a the concept security gateway design of ARINC 429 data bus.



Our *team*



Joshua



Nate



Brandon

Content *overview* ARINC 429

1

ARINC Use Case

2

ARINC Technical
Overview

3

SYSML

4

Threat Model

5

Demo

6

Questions



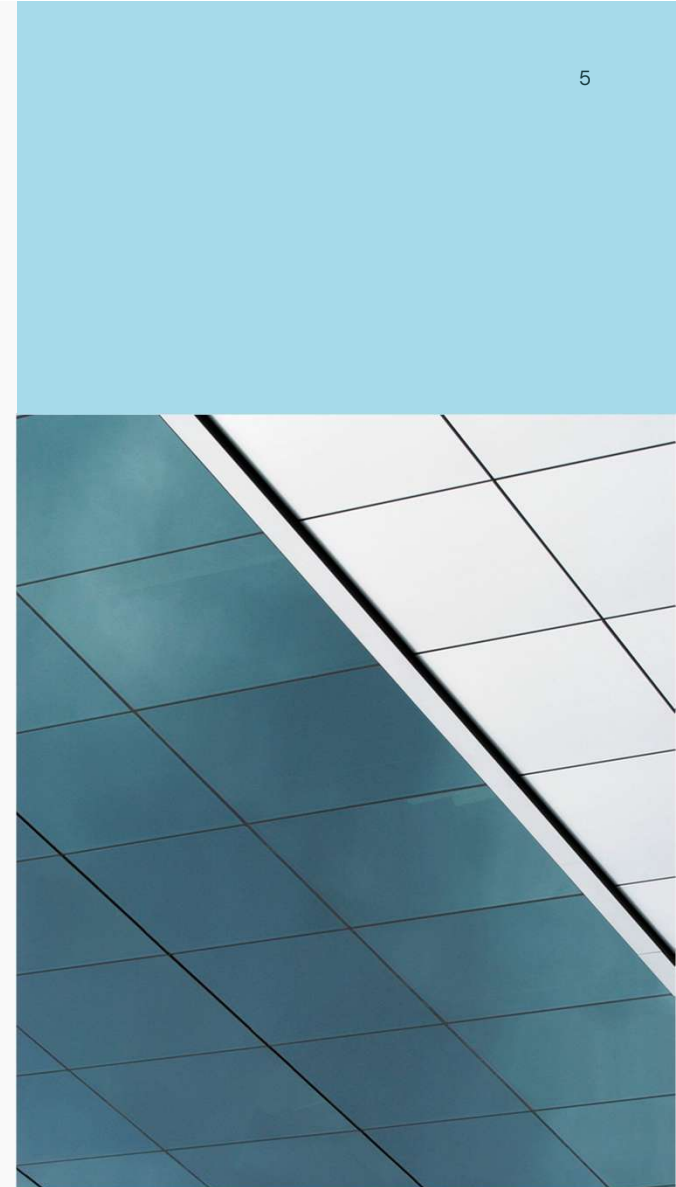
What is ARINC 429?

The ARINC 429 is a data bus used in commercial, private, and military aircrafts to transport avionic data throughout the plane.

This includes all of the flight data like the radar, relative pitch and yaw, airspeed, data to the black box and much more information.

How Does ARINC work?

- 1 transmitter sends data to up to 20 receivers through a single twisted wire pair in which it is unidirectional.
- The LRUs transmits 32-bit words, in which 24 of those bits contain data and 8 bits describe the data.
- There are no addresses for the LRUs but equipment IDs
- Transmission rates can either be 12.5 kHz or 100 kHz with a deviation of 1 and only one of these rates can be used at a time.



How Does ARINC work Continued

32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
P	SSM	Data																		LSB	SDI	Label									

ARINC 429 BNR Word Format

8	7	6	5	4	3	2	1	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
Label								SDI	Data																		MSB	SSM	P		

ARINC 429 Word Transfer Order

P (parity): P bit ensures there are an odd number of 1's. If there are an even number of ones that means the word was damaged in transit.

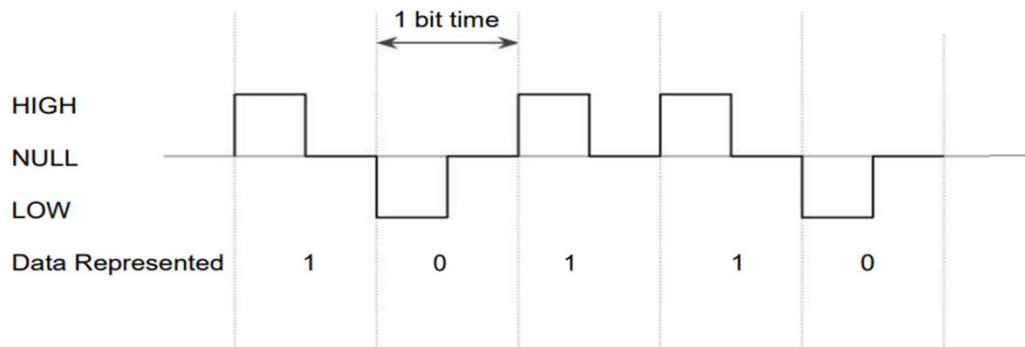
SSM: Provides context for the data (cardinal direction, positive or negative, ect)

Data: The data that the receiving LRU interprets

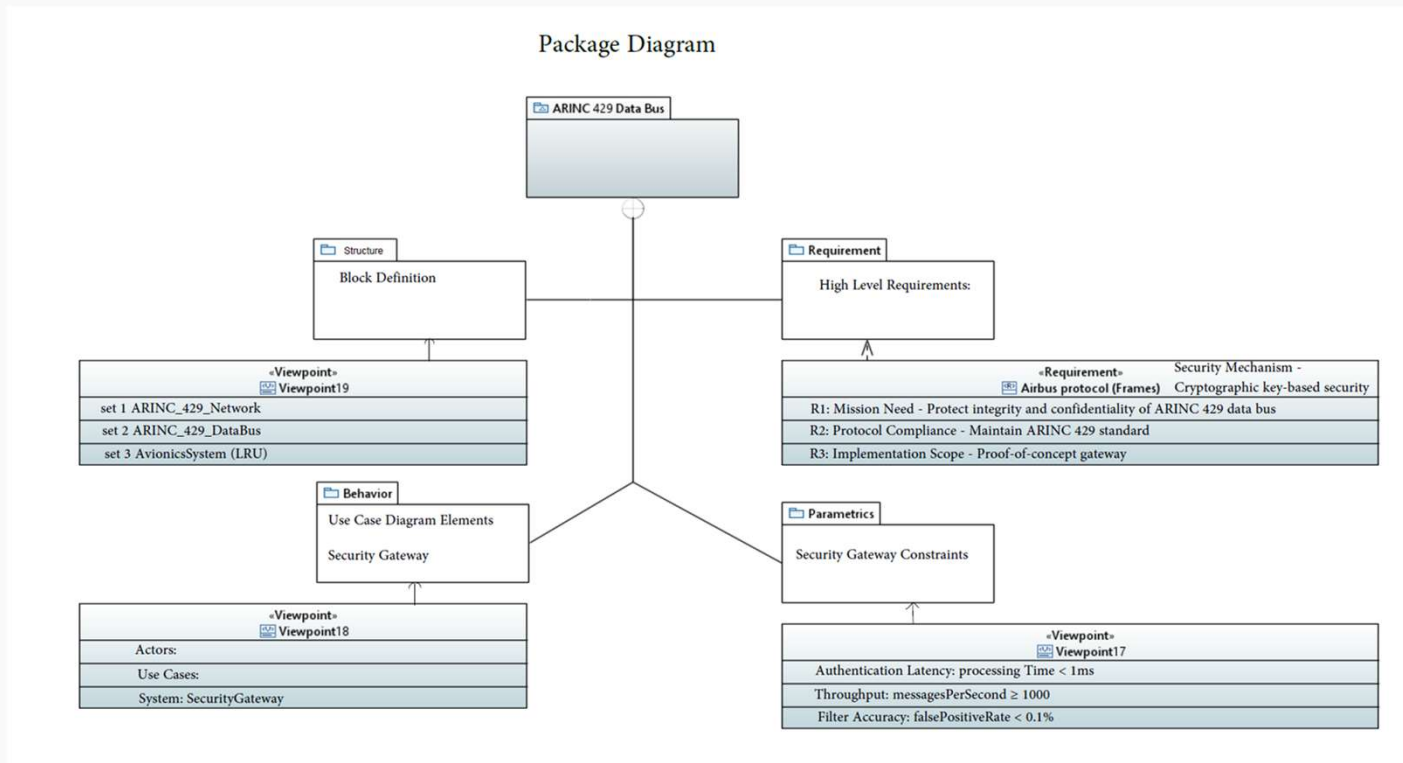
SDI: Informs the LRU of all receivers

Label: Informs LRU on what type of data (velocity, radar, intercom, ect)

TRANSMIT	STATE	RECEIVE
$+10.0\text{ V} \pm 1.0\text{ V}$	HIGH	$+6.5\text{ to }13\text{ V}$
$0\text{ V} \pm 0.5\text{ V}$	NULL	$+2.5\text{ to }-2.5\text{ V}$
$-10.0\text{ V} \pm 1.0\text{ V}$	LOW	$-6.5\text{ to }-13\text{ V}$

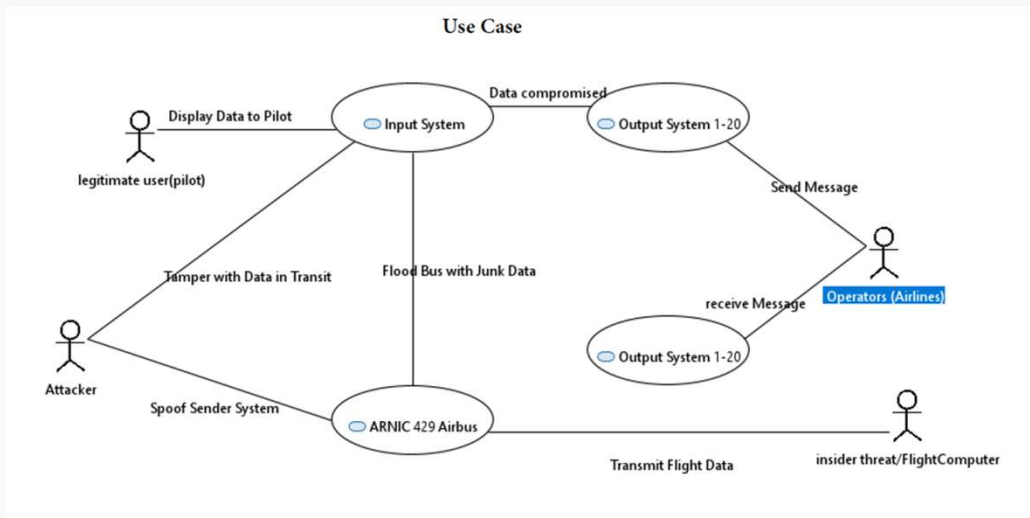


- The data is transmitted in a Return-to-Zero (RZ) format which is a tri state modulation which can has HIGH, NULL, or LOW
- When determining what the data is, the voltages are measured at the output terminal at the source and at the receiver.
- The data is measured in bit cycles where the first half is where the voltage is measured and the second half is where it goes back to NULL.
- Since the cycle returns to zero, there is no need for an external clock



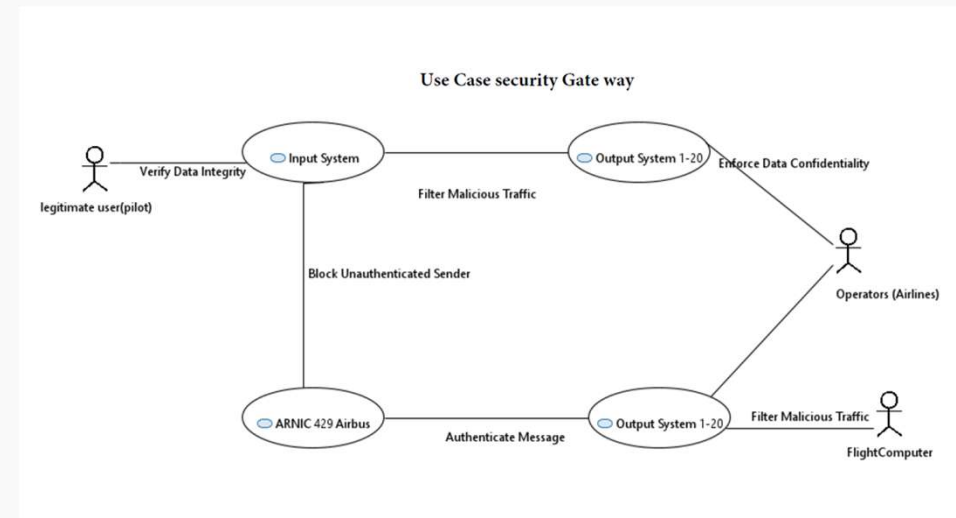
- ARINC 429 data bus is critical domain for aircraft.
- Handles essential data from flight information to radar data to black box recordings.
- Mission is to protect the integrity and confidentiality of data transmitted across this vital bus.

SYSML model (Use case)



Normal Operations:

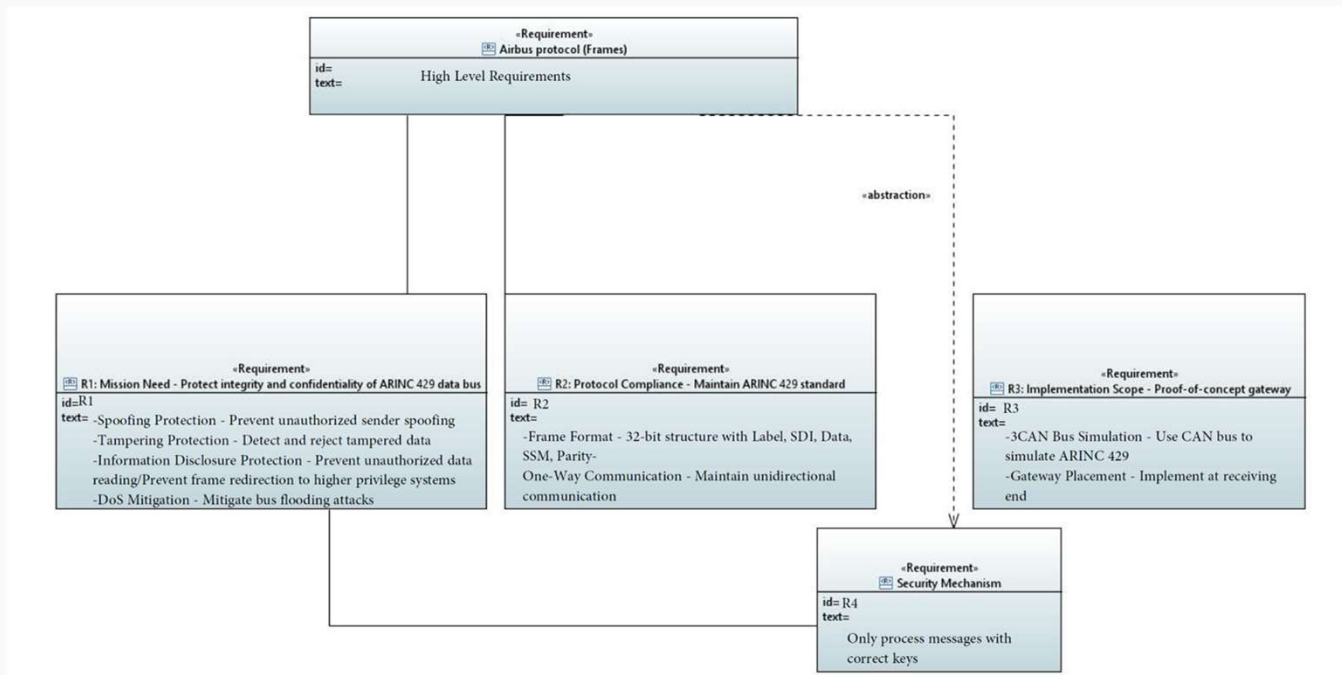
- Transmit flight data
- Display data to the Pilot
- Identified Attack: spoof, Flood Bus...



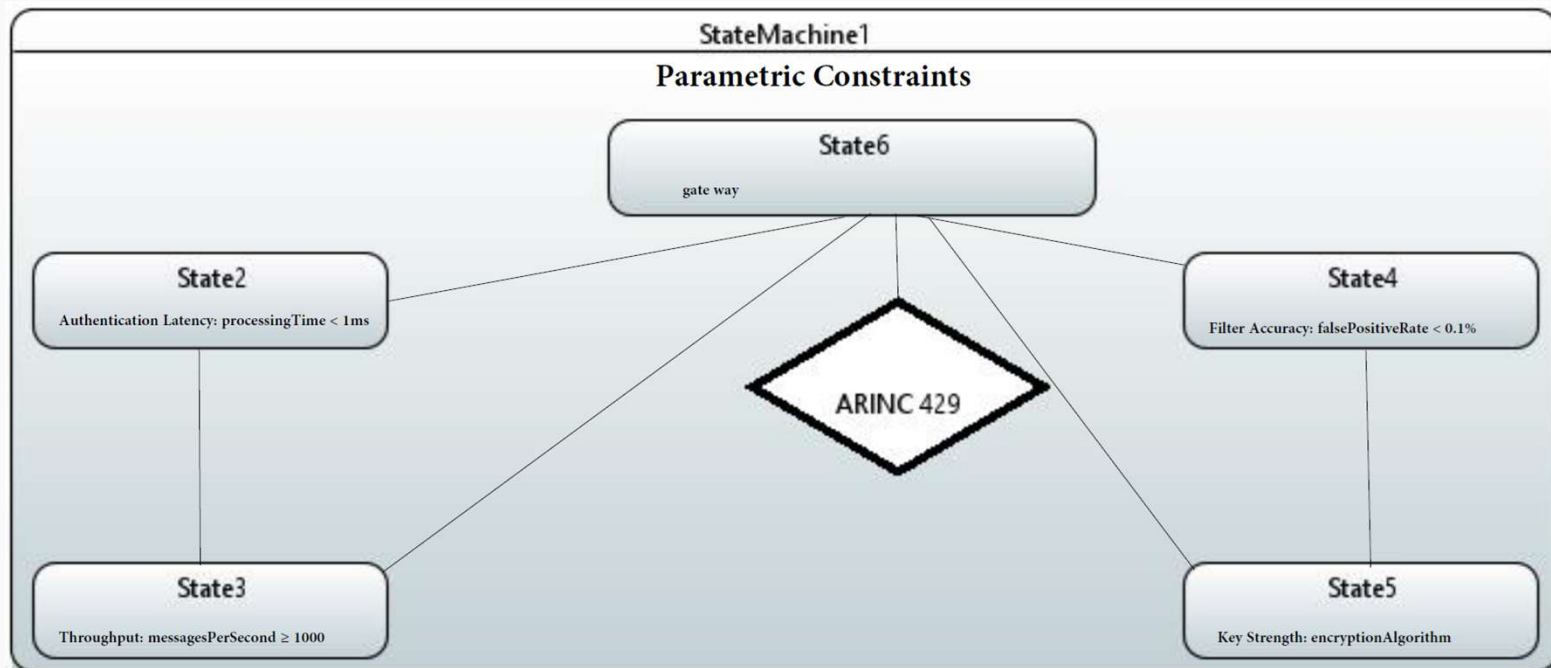
Security Gateway Mitigations:

- Authenticate to verify sender identity.
- Verify data integrity to check for tampering.
- Filter Malicious Traffic to stop bus flooding.
- Enforce Data confidentiality.

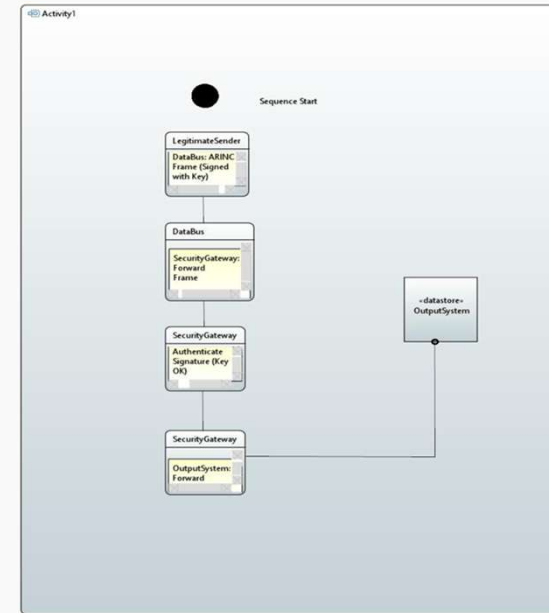
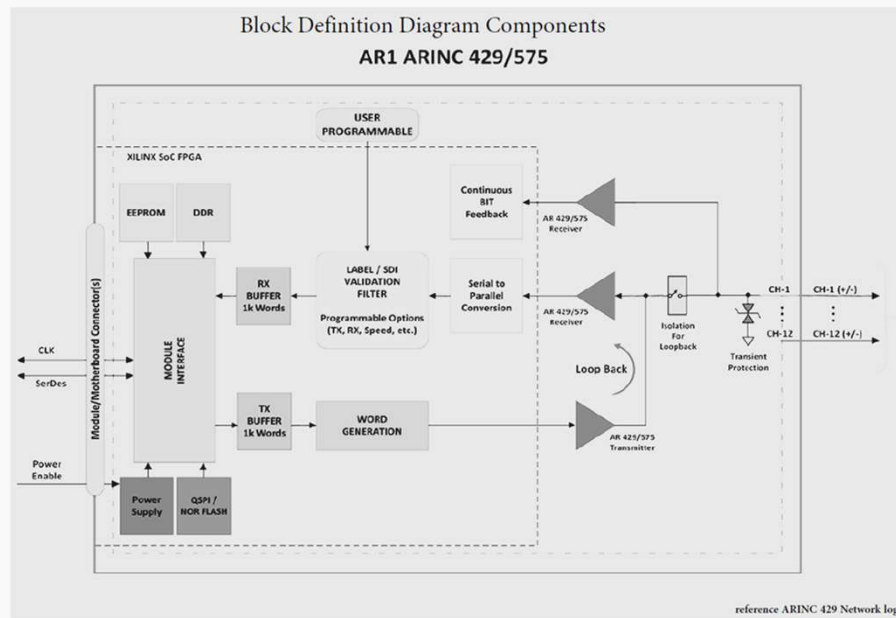
SYSML model (Requirement)



SYSML (Parametric)



Sequence/Block DDC

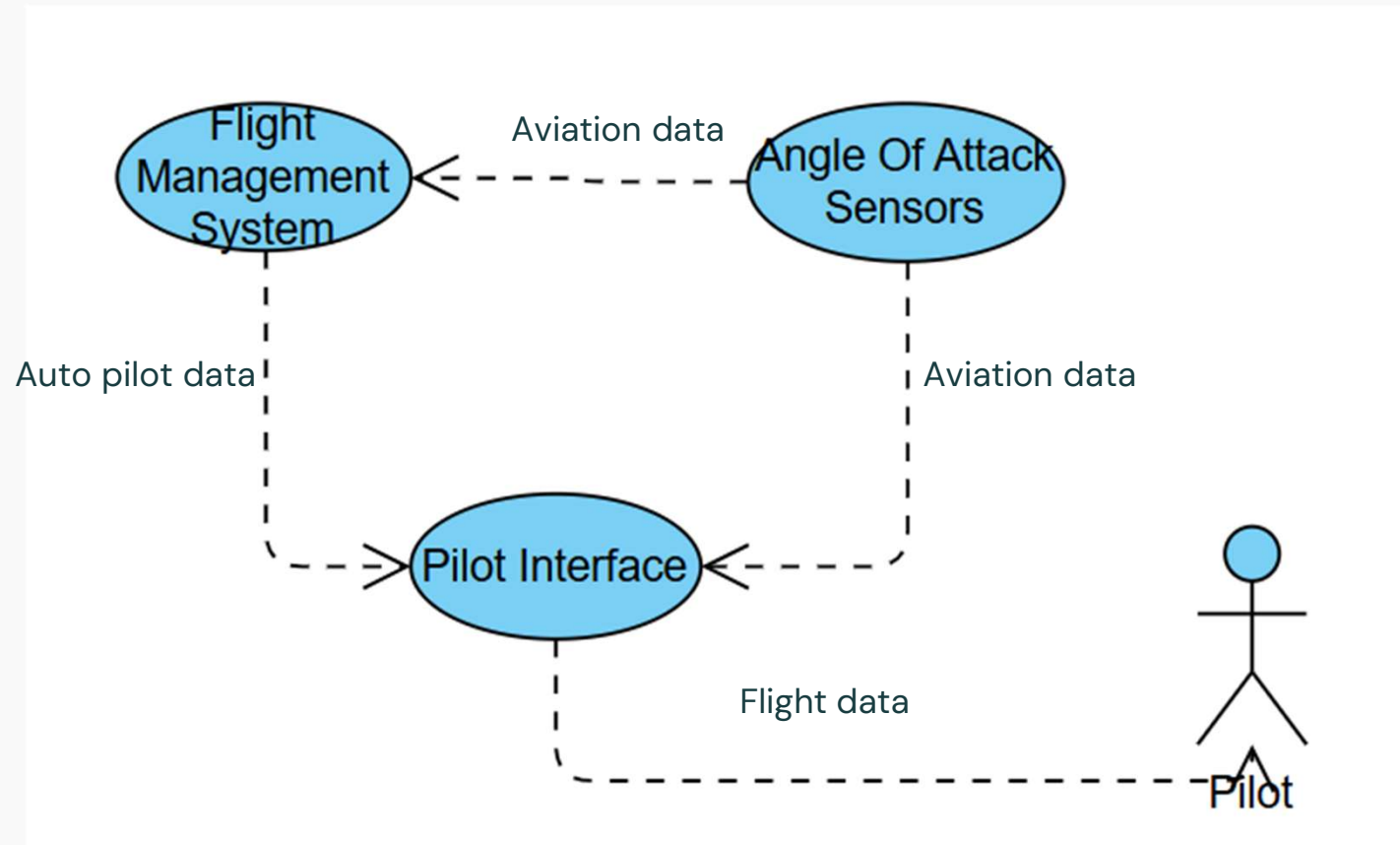


- Normal Secured Operation:
 - Legitimate sender transmits a frame signed with a cryptographic key.
 - The Security Gateway authenticates the signature successfully.
 - Validated data is forwarded to the output system for processing.
- Attack Mitigation Scenario:
 - An attacker transmits a frame with an invalid or missing key.
 - The Security Gateway fails the authentication check.
 - The malicious frame is silently discarded, protecting the output system.

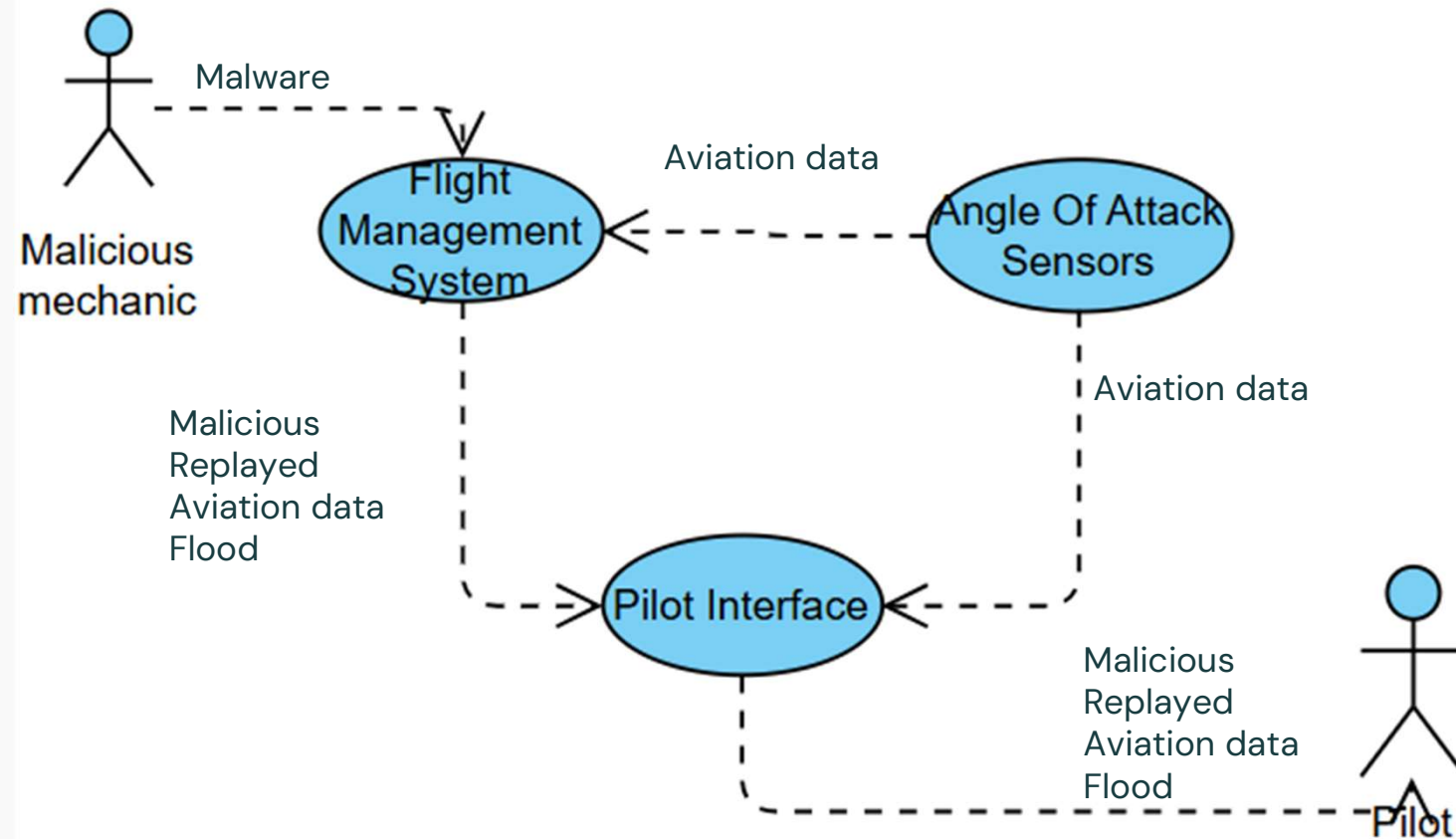
Threat Modeling

	Altered Frames (Tampering)	Replay (Spoofing)
Damage	(3) Can affect all receiving nodes on the same bus	(3) Can affect all receiving nodes on the same bus
Reproducibility	(1) Frame specifications varies by manufacturer	(3) does not require knowledge of data specifications, attack stays the same if the same LRU is compromised each time
Exploitability	(1) Requires physical access, custom malware, and knowledge of how receivers interpret data	(2) Requires physical access and custom malware
Affected users	(3) Potential for all passengers to be affected	(3) Potential for all passengers to be affected
Discoverability	(2) Knowledge of theoretical attack is easily discoverable due to public protocol information, however label and data specifications are not public	(2) Knowledge of theoretical attack is easily discoverable due to public protocol information, however label and data specifications are not public
Total	10 (Medium)	13 (High)

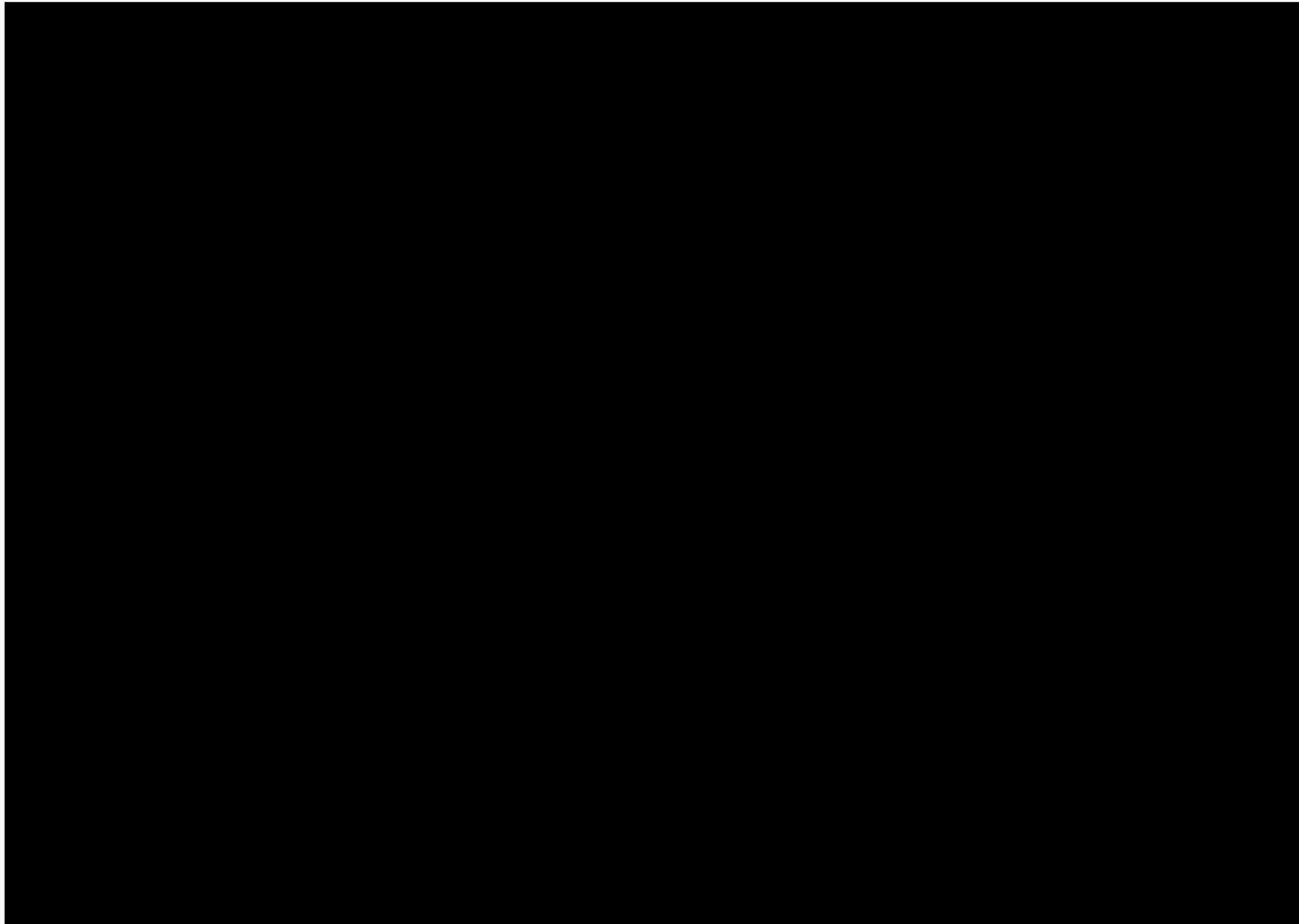
Demo



Demo Cont.



Demo Cont.



References and Questions?

- [1] <https://github.com/JoshuaMFrench/CYSE465-Caterpillars-Group>
- [2] B. Waldmann. ARINC 429 Tutorial, v2.2. (2019). Accessed: September 12th, 2025. [Online]. Available: <https://www.aim-online.com/wp-content/uploads/2019/07/aim-tutorial-oview429-190712-u.pdf>
- [3] <https://www.omg.org/sysml/INCOSE-OMGSysML-Tutorial-Final>



Individual Contribution

Josh

- Initial case study for the proposal with team. .
- Python simulation of communication protocol
- Python simulation of replay attack
- Recorded Demo
- Prepare and structure the powerpoint with team.

Nate

- Initial case study for the proposal with team.
- System Architecture (Sysml model).
- Initial Python simulation for testing communication with terminal.
- Prepare and structure the powerpoint with team.

Brandon

- Initial case study for the proposal with team
- Simulating communication with terminals
- Pushed updates to the Repository
- Prepare and structure the powerpoint with team

Team Effort

It was "All hand on deck" approach for the project until the Midterm.

Thank you